

KEDAR DAMALE

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Portfolio · GitHub · LinkedIn

SUMMARY

Applied Data Scientist with nearly 1.5 YOE building end-to-end AI/ML solutions across fintech, financial automation, pharma, and medical research. Skilled in Data Science, Machine Learning, Generative AI, multi-agent systems, model development, data pipelines, and production deployment, with experience taking solutions from research and experimentation to production-ready systems.

SKILLS

- **Foundational Mathematics** : Linear Algebra, Calculus, Probability and Statistics
- **Programming and Data** : Python, NumPy, Pandas, Polars, SQL
- **Generative AI** : LangChain, LangGraph, Multi-Agent Systems, RAG, HITL, MCP
- **Machine Learning** : Scikit-learn, XGBoost, Feature Engineering, Model Evaluation, Hyperparameter Tuning
- **Deep Learning** : PyTorch, CNNs, YOLOv8, Transfer Learning, Model Training and Evaluation
- **DevOps** : FastAPI, Docker, AWS, MLflow, Git/GitHub, CI/CD, ETL/ELT Pipelines

WORK EXPERIENCE

ML and GenAI Engineer

Globalspace Technologies Ltd.

Jan 2026 – Present
Navi Mumbai, MH, India

1. Budget Automation & Incentive Calculator

- The existing budget and incentive workflow maintained a separate execution tree for each Medical Representative (MR), which became difficult to manage beyond roughly 20–30 employees and could not scale to hierarchies of thousands.
- Designed a unified hierarchy model and dynamic scoring engine that allocates MRs as they join or leave, with human-in-the-loop policy configuration and automatic rebalancing after edits.
- Replaced the per-MR structure with one scalable model, making incentive planning workable across large teams and allowing business users to adjust policies without specialist schema or admin support.

2. PatGPT

- Searching reference material alone could not answer commercial planning questions across more than 3 million pharmaceutical sales and tender records or help teams evaluate possible actions.
- Built PatGPT as an agentic planning assistant for forecasting, simulation, prediction, and historical analysis, with human review checkpoints and answers grounded in supporting evidence.
- Delivered a production-ready workflow with streamed responses, safe retry handling, and checksum-verified data promotion and rollback, supporting more reliable analysis and release operations.

Freelance Automation Developer

Propelligence Advisors Pvt. Ltd.

May 2025 – Jan 2026
Thane, MH, India

- Reconciling more than 10,000 purchase-register (PR) and GSTR-2B invoices took five to six days per cycle because vendor and invoice details differed and matches needed manual validation.
- Engineered an automated reconciliation pipeline using RapidFuzz fuzzy matching (80%+ threshold), Polars processing, and constant-memory report generation to handle large files efficiently.
- Reduced reconciliation time by more than 99.9% and contributed the module to Granska, a five-module audit automation suite adopted by two CA firms and about 20 auditors.

PROJECTS

1. Chessablanka – Chess Position Analyzer | GitHub

Apr 2025 – Jan 2026

- Analysing a chess position from a photograph traditionally requires manually transcribing the board before an engine can evaluate it, making the process slow and error-prone.
- Built an end-to-end computer-vision pipeline that detects the board and maps its grid using a custom YOLOv8 model trained on 700+ images and augmented to 1,700 with Roboflow.
- Achieved 98.57% detection accuracy across 500+ test images and connected recognized positions to Stockfish for immediate evaluation and best-move analysis.

2. NeuroTRIBE – Cortical Response Analysis for ADHD Research | GitHub

Aug 2026 – Present

- Testing whether ADHD is associated with differences in stimulus-evoked cortical responses requires a reproducible comparison against an appropriately matched baseline.
- Built a TRIBE v2 analysis pipeline over Healthy Brain Network movie-fMRI data to compare participants with confirmed ADHD against a matched cohort.
- Added Docker Compose orchestration to make cohort-level exploratory analysis repeatable while keeping legally gated inputs outside the workflow and non-blocking.

3. Cattle Monitoring using DBSCAN and ESP32 | GitHub

Oct 2025 – Nov 2025

- Farmers monitoring free-grazing cattle had to rely on manual rounds, making it hard to see herd locations and recurring grazing patterns in real time.
- Led a team building a distributed tracking system with one ESP32 master and multiple child nodes that send GPS and RSSI readings to a Flask and MongoDB backend.
- Applied DBSCAN to identify grazing zones and displayed the resulting locations on a live Next.js dashboard, giving farmers a clearer view of herd movement.

EDUCATION AND ACHIEVEMENTS

- **Qualified GATE** — Data Science and Artificial Intelligence (DA) paper; 8.5/10 CGPA in Information Technology Engineering, University of Mumbai
- **Published Automation Suite for CA & Audit Firms** — Adopted by 2 CA firms and used by about 20 auditors